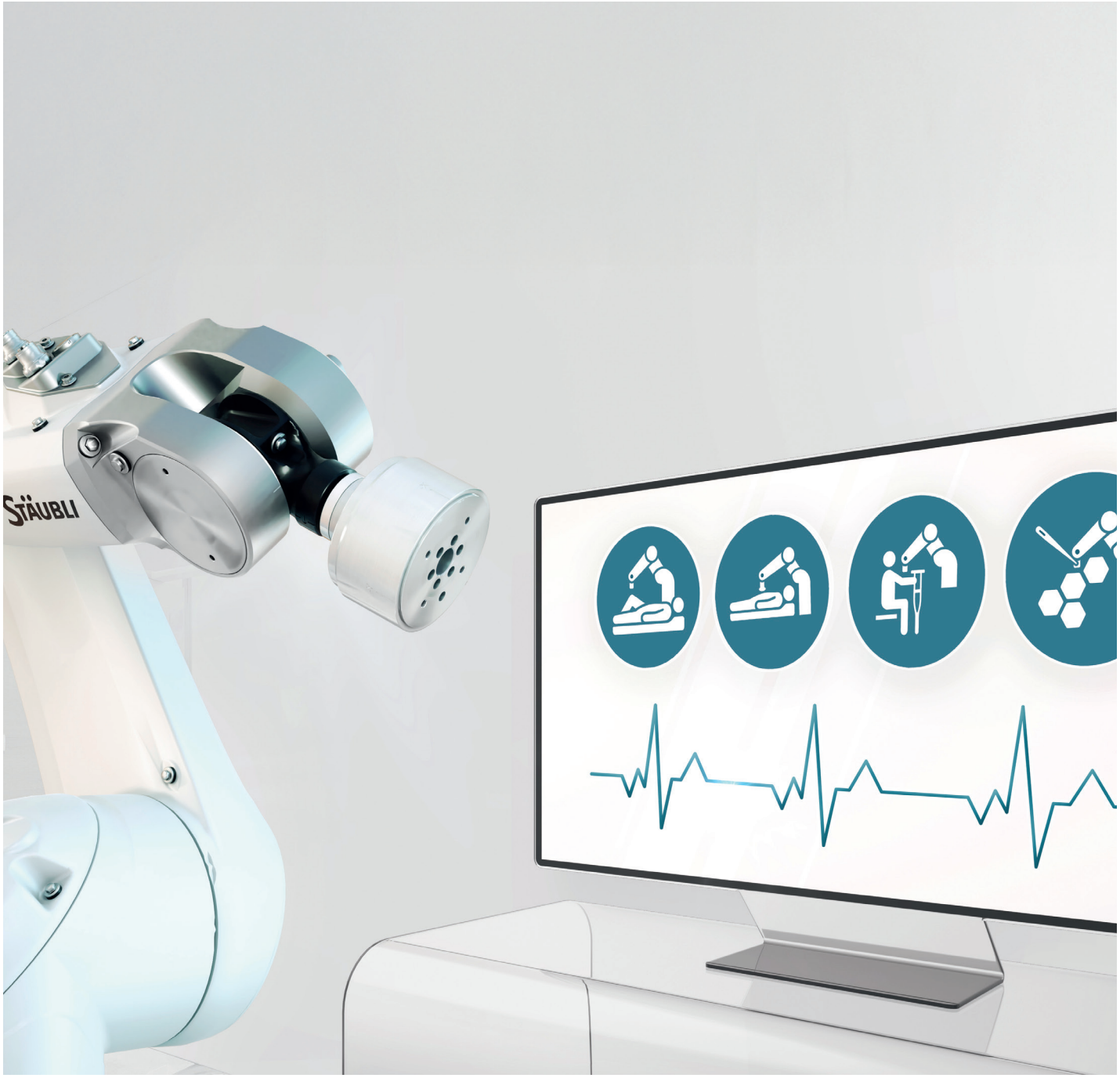
FAST MOVING TECHNOLOGY

Patient care applications

Robotics | Experts in Man and Machine



STÄUBLI IN LIFE SCIENCES

Robots contributing to human well being

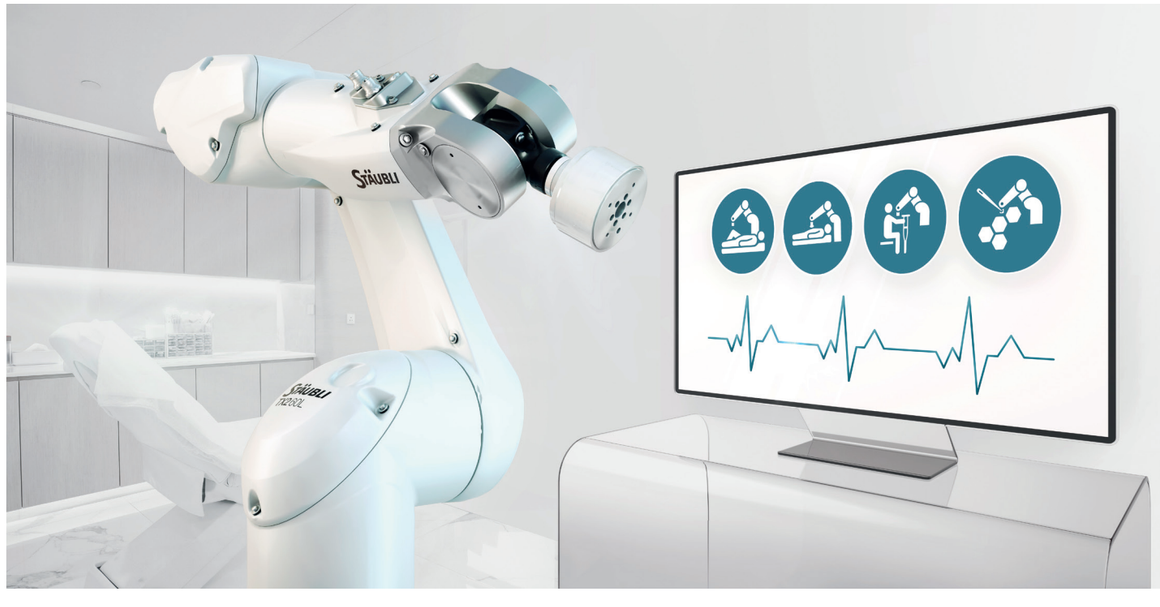
For many years now, Stäubli Robotics has been on a mission to develop robots for life science applications. Stäubli robots are setting the benchmark for hygiene and safety.

Whenever accuracy has to be maintained alongside factors such as particle emissions, easy-to-clean surfaces and availability, Stäubli has long led the way.

Stäubli's unique robot portfolio, decades of industry experience and strong customer focus make it a valued partner for renowned life science companies around the globe.

Life Science

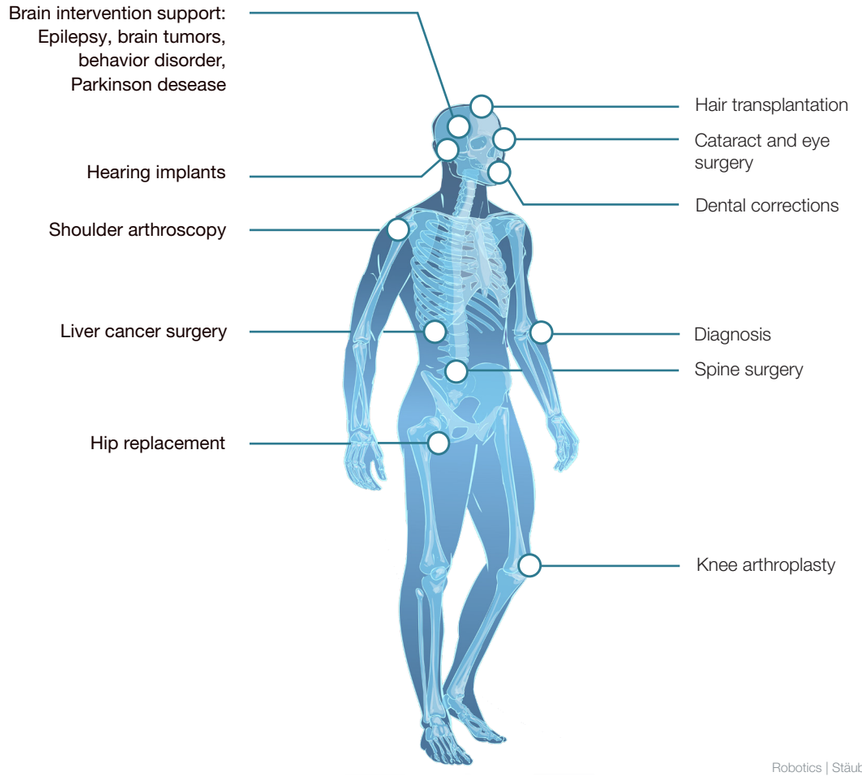
- **Laboratory research, drug discovery and diagnostics**
- **Pharmaceuticals and biotechnology**
- **Manufacturing of medical devices**
- **Hospital automation**
- **Patient care**



MORE THAN 30 YEARS OF EXPERIENCE

Pioneer in robotics patient care

- **1985** — Stäubli **PUMA 560 robot**: Brain biopsy assistance (needle placement, scanner support)
- **1998** — **Ortho Maquet**: Hip and knee surgery
- **2008** — **Stericlean range**: Launch of dedicated range for medical and pharma applications
- **2013** — **Medtech**: Spine and brain surgery
- **2018** — **Medtech Zimmer Biomet**: Knee surgery
- **2019** — **Keranova**: Cataract operation
- **2020** — **BHS Technologies**: Surgery assistance (precision vision positioning)
- **Quantum Surgical**: Partnership for liver surgery



DESIGNED FOR PATIENT CARE

Medical package

Stäubli has developed a unique offer to meet a number of constraints present in patient care. Therefore, the Stäubli medical arm is specially adapted to be the perfect partner in patient care.

As in our industrial solutions, the medical range ensures consistent performance and reliability covered by worldwide Stäubli support and service.

Stäubli currently operates in 29 countries, with agents in 50 countries to provide comprehensive solutions with long-term support.

Safety

Safety performance level SIL3 PLe

Hygienic Design

Complete range of clean, cleanable and up to sterilizable robots

Aesthetics

Extreme smooth finishing

Due to a unique set of modular SIL3-PLe safety functionalities (safe stop, safe speed, safe zone, ...) TX2/CS9 systems are designed for collaborative applications with the highest safety level.

On top of that, reliable stopping/braking performance is ensured by system embedded brake test.

- Up to ISO class 2
- Good Manufacturing Practices (GMP) grade A
- Complete Stericlean range of H_2O_2 compliant robots available
- Protection class up to IP67
- H1 oil optional
- Optional FDA certified wrist coating

- Smooth, white, high quality finishing with specific surface preparation
- Specific wrist treatment for increased robustness (available in grey/ aluminum or black)

DESIGNED FOR PATIENT CARE

Your partner for safer intervention

Clean and hygienic

Fully encapsulated kinematic designed under hygienic aspects

Unmatched precision

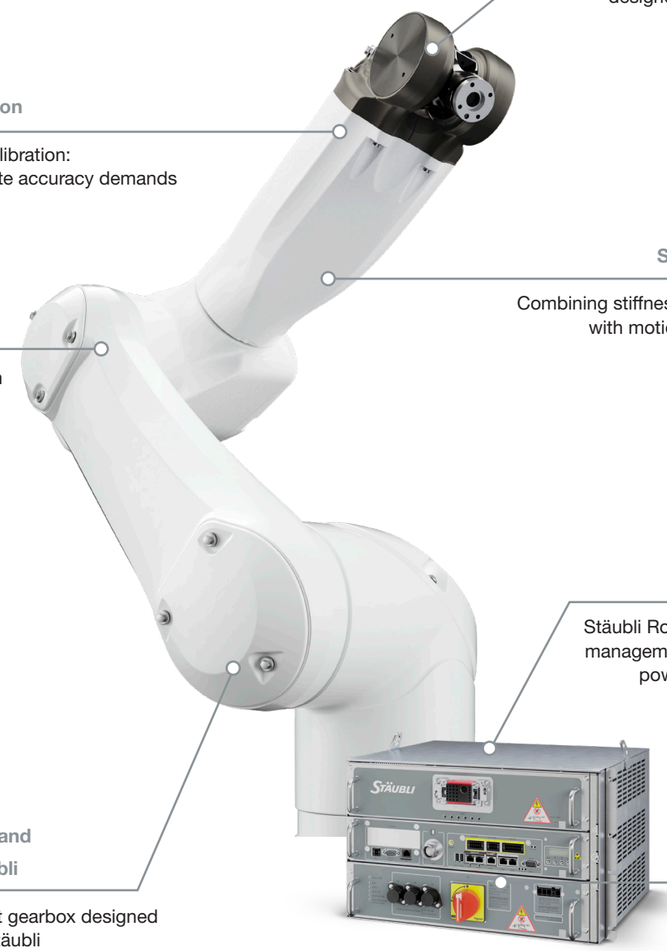
Optional absolute calibration:
For extended absolute accuracy demands

Safe motion expert

Combining stiffness and performance with motion control expertise

Personalized

Optional color customization
Visual personalization
Advanced customization



Software / Application development environment

Stäubli Robotics Suite simplifies project management by giving access to all the powerful controller functionalities

System designed and produced by Stäubli

Patented JCS smart gearbox designed and produced by Stäubli

Controller

Cleverly connected to your world

WIDE RANGE TO MEET YOUR NEEDS

High reliability of your solution

Consistency

Consistent performance and quality
Additional parameters related to Aesthetics, Performance and Behavior are checked, controlled and documented during production of your medical robot for your quality and traceability requirement.

Product Change Management (PCM)
Assuring a reliable development and life-time control and securing your investment due to regular and in time product change notifications. PCM is available for Stäubli hardware and software.

Connectivity

Open architecture
Any system data accessible and linkable to all interfaces, including OPC UA.

Network Time Protocol (NTP)
Date/time synchronization of local equipment network with a time reference.

CS9-EP (External Power)
Alternative choice for even more compact and flexible integration for optimizing the power supply architecture of your OEM system.

Performance

High performance controller
Easy to integrate into different systems and peripheral equipment.
Huge bandwidth of powerful controller functionalities offered by Stäubli Robotics Controls.

Stäubli Robotics Suite
Easily enables the realization of individual customer needs for today's smart production.
Encryption of your code allows protecting your application development (VAL 3).

A comprehensive range available

MODEL	**TX2**-40 Medical package	**TX2**-60 / 60L Medical package	**TX2**-90 / 90L / 90XL Medical package	**TX2**-140 Medical package	**TX2**-160 / 160L Medical package
Maximum load Maximum load under conditions	2 kg 2.3 kg	4.5 kg / 3.7 kg 9 kg / 5 kg	14 kg / 12 kg / 7 kg 20 kg / 15 kg / 9 kg	30 kg 34 kg	30 kg / 20 kg 34 kg / 28 kg
Nominal load	1.7 kg	3.5 kg / 2 kg	6 kg / 5kg / 4kg	20 kg	20 kg / 14 kg
Reach (between axis 1 and 6)	515 mm	670 mm / 920 mm	1000 mm	1510 mm	1710 mm / 2010 mm
Repeatability - ISO 9283	± 0.02 mm	± 0.02 mm	± 0.02 mm	± 0.015 mm	± 0.02 mm/ 0.03 mm
Weight	29 kg	52.2 kg / 52.5 kg	114 kg / 117 kg / 119 kg	250 kg	260 kg / 265 kg
Sealed cable option	Horizontal or vertical				
Multiple mounting positions	360°				

ALWAYS CLOSE TO YOU

A team committed to support you

Conception

Our team of experts will assist you to evaluate your concepts and take your first steps

- Full line of training courses: Stäubli offers a range of dedicated training courses intended for all of the different users of your systems (maintenance technicians, programmers and operators).
- Privileged application development package: combining robot supply and technical expertise

Simulation and setup

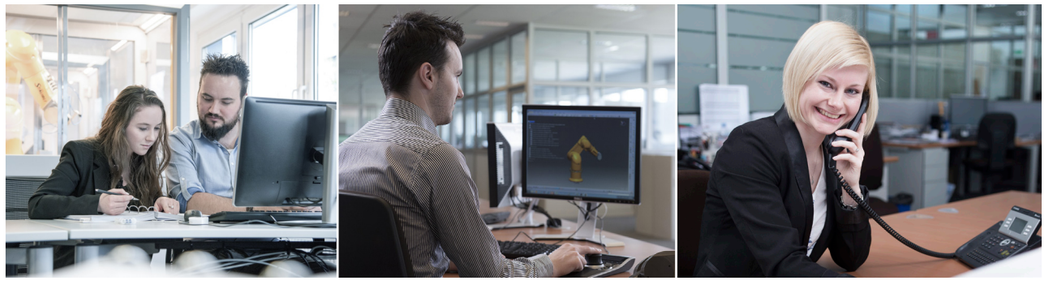
Our Sales and Application/Feasibility engineers work to bring your project to life through simulations and tests.

- Workspace / Feasibility studies
- 3D visualization
- On-site programming assistance
- Real-time testing
- Direct access to dedicated R&D teams

Operation, improvement and maintenance

After Sales service that fulfills your needs, with the goal of delivering unrivalled system availability.

- Robot built on the basis of a proven and reliable industrial robot that ensures the availability of spare parts throughout the country.
- On-site intervention: Globally we intervene on-site in 48 hours with an average waiting time of less than 24 hours.
- Dedicated hotline to support your applications development or troubleshooting.
- Preventive maintenance program to ensure the continuous high performance of your equipment and its lifetime.



Robots at the heart of clinical interventions

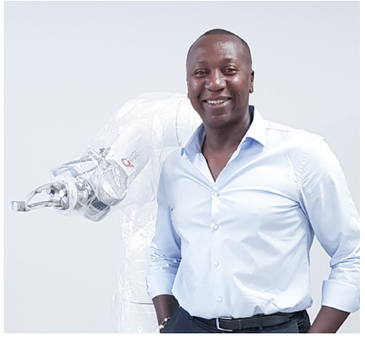
A win-win partnership strategy

Quantum Surgical and Stäubli, both European leaders in their respective fields, are joining forces to create a Centre of excellence in oncology robotics.

Co-founded in 2017 by Bertin Nahum, Quantum Surgical is an innovative Robotic Surgery Start-up company that is aiming to conceive an innovative solution; a combination of powerful software and robotic platform for minimally invasive therapies in liver cancer.

This platform combining robotics, image analytics, and artificial intelligence will allow more and more patients to benefit from safer interventional oncology procedures, especially in the case of early-stage liver cancers. The incidence of liver cancer in the population is, unfortunately, growing very rapidly. Liver cancer is one of the most common cancers with more than 800,000 cases each year in the world.

This strategic partnership aims to combine this savoir-faire to strengthen the role of robotics in minimally invasive cancer treatments and to build a world leader.



"We are honored to work with Stäubli, a company recognized for its technological excellence and savoir-faire, in order to share a vision of robotics and automation in healthcare"

Bertin Nahum,
President, and Founder of Quantum Surgical

Founded in 2017, Cyber Surgery is a start up in San Sebastian (Spain) who develops a robotic system for spinal procedures.

The solution guides the surgeon in order to make the procedure safer, more accurate and minimally invasive.
The three main benefits are:

- Reinterventions reduction due to the accuracy provided by the robot,
- Postoperative hospital stay time shortening because the surgery is performed in a minimally invasive way producing less damage to soft tissues and bleeding,
- Reduced X-ray exposure thanks to the fact that the robot provides a guide to the surgeon.

Due to Stäubli's leadership in the medical robotics fields and the good relation between the two companies, Cyber Surgery has been in direct contact with Stäubli's salespeople and R&D engineers since the very early stages of the project. From the development point of view, Stäubli has always provided a very reliable support to help the engineers integrate their systems in the solution.

Andres Amarillo, Cyber Surgery CTO explains: "Reliability, good customer service and product quality are strong points of Stäubli that help speed up developments when working in a competitive environment like ours."

Andres Amarillo,
Cyber Surgery CTO

WORLD FIRST ROBOTICSCOPE

Seeing more clearly than ever before

TX2-60L robot manages the precise and jerk-free motion sequences of the RoboticScope.

A high-precision robot, a high-resolution 3D camera and an innovative head-mounted display – the world's first RoboticScope has arrived on the market.

The RoboticScope was developed in 2016 by BHS Technologies GmbH in Innsbruck. The success of BHS in realizing these objectives is demonstrated by the functionality of the RoboticScope. The surgeon looks directly into the operating field via a Head-Mounted-Display (HMD).

The two digital microdisplays integrated into the HMD directly in front of the surgeon's eyes show high-resolution, real-time 3D images relayed by the dual camera lenses mounted on the robot head. The position of the robot determines the visible image, allowing the surgeon to operate in a freely selectable and completely relaxed head position.

The surgeon retains control over the exact image field and viewing angle, automatically steering the robot with contact-free head movements.

Using this method, the surgeon can select the viewing angle of the 8x zoom lenses and even control a sophisticated menu from a microdisplay, allowing the activation and control of further functions.

Some 500 surgeons around the world have now trialed the RoboticScope. The device is so intuitive that more than 90 percent of users master it after just ten minutes. This is an incredible result that was completely independent of the age of the test candidates: the "digital generation" scored no better than the experienced veterans.

The technology behind the RoboticScope
A high-precision Stäubli TX2-60L robot delivers the precise and jerk-free motion sequences of the RoboticScope. As with the entire range of latest 6-axis robots, the enclosed structure allows for a protection class IP65, with the wrist being IP67 compliant. With cleanroom class ISO 5 as a standard, the robot allows BHS to also save on the potential extra costs of upgrading to a cleanroom version.

In addition, the new machines are more lightweight, rigid and powerfully motorized, making them even more dynamic and, with an accuracy of 0.02 millimeters, more precise than ever before.

The robot is maneuvered by a Stäubli CS9 controller as well as its own digital safety encoders (one per axis) and an integrated safety board. All safety functions meet the strict requirements of the SIL3/PLe category. In order to ensure the highest level of safety, every movement of the robot is monitored built-in sensors. For this purpose, all coordinates of the robot as well as speed and acceleration, are controlled and logged in real time. These functions of the highly dynamic robot make it perfect for MRC (man-robot collaboration) scenarios, in this case with surgeons.

ROBOTIC SURGICAL ASSISTANT

The pursuit of perfection: robots in the OR

ROSA® technology is an FDA-approved, next-generation surgical robot in use at more than 200 hospitals. Developed by surgeons for surgeons, Zimmer Biomet's robotics technology, complete with an integrated navigation platform, has assisted in numerous surgeries around.

The ROSA® technology system consists of two mobile units that are placed adjacent to the operating table. One is used for the Spine and Brain Application, and another for the Spine Application only. One of the two units has a camera to continuously monitor the motion of the other unit in three dimensions. A Stäubli six- axis robot is mounted on a rigid frame. The robot arm, holding the instrument, carries out its movements in parallel with those shown on the monitor. A distinctive feature of the system is that it recognizes and takes into account the slightest shift of position, for example of the thorax during breathing. This dynamic tracking is designed for constant adjustment of the parameters, allowing for the precision of the instrument trajectory and minimizing procedural risks.

Combining innovative robotics with surgery, ROSA® technology can also incorporate advanced implants made from innovative materials. Combining advanced imaging methodologies is another reason for robotics in the OR. ROSA® Knee system imaging modalities, combined with intraoperative data, allows surgeons to conduct a "virtual knee replacement" with predictive outcomes of balance and implant placement before performing any cuts.

Six degrees of freedom and meticulous monitoring
In terms of mechanics, ROSA® consists of a robotic arm with six degrees of freedom. Similar to the human arm, it can perform exact multidimensional movements. This makes it a high-precision aid for surgeons who require maximum freedom of movement in every imaginable plane in order to carry out procedures to the highest standards. The mechanical components are complemented by assisted navigation. This unit allows surgeons to visualize their instruments in real time on patient images. The robot's haptic capabilities allow surgeons to manually and precisely guide and manipulate their instruments within the limits set in the planning phase. In this human-machine interaction, surgeons can continue to apply their individual skills while the system learns from them.

A milestone in modern surgery – robotic surgical assistants.

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● Stäubli Units ○ Representatives / Agents

Global presence of the Stäubli Group

www.staubli.com

Non-contractual document. The examples presented are no guarantee of the suitability of Staubli robots to meet specific medical requirements. The characteristics of the products are given as an indication only, shall not constitute a guarantee and may be subject to modification without any notice.

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